

Database Schema

DynamoDB tables, indexes, access patterns and samples

The complete data model for the platform: four DynamoDB tables, their partition/sort key designs, global secondary indexes, the access patterns each index serves, and realistic sample items.

Document	Database Schema
Date	25 June 2026
Audience	Executive summary + engineering detail (layered)

Status

Confidential — internal

1. Overview & conventions

Storage is Amazon DynamoDB. The design is access-pattern driven: each table uses composite keys (PK + SK) and a small set of global secondary indexes (GSIs) to serve the queries the application needs without scans on the hot path. Single-table-style overloading is used within each table (multiple entity kinds share a table, distinguished by key prefixes).

- PK = partition key, SK = sort key.
- Item kinds are namespaced by key prefixes (e.g. EMG#, VEH#, HOSP).
- SK = "META" denotes the primary item; child rows (events, rides, fuel) use other SK prefixes.
- On-demand capacity mode — no provisioned throughput to manage.

2. Tables at a glance

Table	Holds	PK / SK pattern
ReferenceData	Locations, zones, hospitals, fire stations, policy	PK = type (LOC/ZONE/HOSP/FIRE/POLICY), SK = id
Fleet	Vehicles, drivers, allotments, fuel logs	PK = VEH#/DRV#/ALLOT#..., SK = META/FUEL#...
TransportRequests	Emergencies, requests, bookings + audit events	PK = EMG#/REQ#/BK#<id>, SK = META / EVT#<ts>
ShuttleCards	Shuttle entitlement cards and rides	PK = CARD#<id>, SK = META / RIDE#<date>#<id>

3. ReferenceData

Reference/master data, partitioned by record type so each type is a single-partition query.

3.1 Schema

Attribute	Type	Notes
PK	S	LOC ZONE HOSP FIRE POLICY
SK	S	The record id (e.g. hosp-tmh, zone-bistupur)
id	S	Same as SK, surfaced for the app
lat / lng	N	Coordinates (LOC/HOSP/FIRE)
zone_id	S	Owning zone
specialties / capability	L / N	HOSP only — case types handled, capability tier
ref / polygon / color	M / L / S	ZONE only — centroid, boundary, map colour
levels / params	L / M	POLICY only — operational parameters

3.2 Access patterns

Pattern	How
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List all locations / zones / hospitals / fire stations	Query PK = "LOC" / "ZONE" / "HOSP" / "FIRE"
Latest policy	Query PK = "POLICY", newest first, limit 1
Get one hospital/zone by id	GetItem PK=type, SK=id

3.3 Sample item (hospital)

```
{ "PK":"HOSP", "SK":"hosp-tmh", "id":"hosp-tmh",
  "name":"Tata Main Hospital", "lat":22.7868, "lng":86.1958,
  "specialties":["Cardiac","Trauma","General","Pediatric"],
  "capability":5, "zone_id":"zone-bistupur" }
```

4. Fleet

Vehicles and drivers (plus allotments and fuel logs), with GSIs that let the dispatcher find a free unit near a zone in one query.

4.1 Schema

Attribute	Type	Notes
PK	S	VEH#<id> DRV#<id> ALLOT#<emp>
SK	S	META (entity) FUEL#<date>#<id>
type / status	S	ambulance firetruck bus car ; idle enroute maintenance
home_zone_id / driver_id	S	Vehicle home zone; bound driver
reg / fuel / odometer	S/N/N	Registration and telemetry
GSI1PK / GSI1SK	S	ZONE#<zone>#VEH / <status>#<type>#<id>
GSI2SK / GSI3PK	S	driver status index / VEHSTATUS#<status>

4.2 Indexes & access patterns

Index	Serves
GSI1-zoneveh (PK ZONE#<zone>#VEH)	Find nearest idle unit of a type: begins_with(GSI1SK, "idle#<type>#")
GSI2 (driver status)	List drivers by availability
GSI3 (VEHSTATUS#<status>)	Fleet utilisation by status
Scan SK = META	Full fleet snapshot for the console

4.3 Sample item (vehicle)

```
{ "PK":"VEH#veh-sonari-amb-1", "SK":"META",
  "id":"veh-sonari-amb-1", "type":"ambulance", "status":"idle",
  "reg":"JH05-AM-1112", "home_zone_id":"zone-sonari",
  "driver_id":"drv-sonari-amb-1",
  "GSI1PK":"ZONE#zone-sonari#VEH", "GSI1SK":"idle#ambulance#veh-sonari-amb-1" }
```

5. TransportRequests

The operational ledger: emergencies (EMG), transport requests (REQ) and shuttle bookings (BK), each with a META item plus append-only EVT# audit rows. Status/zone/source/vehicle GSIs support the dispatch board and queue handling.

5.1 Schema (emergency)

Attribute	Type	Notes
PK / SK	S	EMG#<id> / META (and EVT#<ts> for audit)
entity / status	S	EMG ; EN_ROUTE QUEUED NO_HOSPITAL COMPLETED CANCELLED
kind / case_type / severity	S	medical fire blood ; specialty ; Critical Urgent Normal
pickup / pickup_zone_id	M / S	Location ref or lat/lng; nearest zone
assigned_vehicle_id / driver	S	Dispatched unit and driver
hospital_id / fire_station_id / blood_bank_id	S	Destination by kind
distance_km / eta_min / eta_to_pickup_min	N	Traffic-aware ETA fields
eta_complete / traffic_factor	N	Completion epoch; congestion multiplier
incident_id / patients_count	S / N	Mass-casualty grouping
contact / track_token	M / S	Notify target; live-tracking secret
GS12/3/4/5	S	status, zone, source, vehicle indexes

5.2 Indexes & access patterns

Index	Serves
Scan SK = META (+ filter)	Board list of all emergencies/requests/bookings
GetItem PK=EMG#<id>, SK=META	Single incident status / tracking
GS12 (entity#STATUS#<status>)	Queued/active by status, severity-ranked
GS13 (ZONE#<zone>)	Incidents by zone
GS15 (VEH#<id>)	Trips by vehicle

5.3 Sample item (emergency)

```
{ "PK":"EMG#EMG-784", "SK":"META", "entity":"EMG", "id":"EMG-784",
  "kind":"medical", "case_type":"Cardiac", "severity":"Critical",
  "status":"EN_ROUTE", "pickup":{"lat":22.8145,"lng":86.2207},
  "assigned_vehicle_id":"veh-kadma-amb-1", "hospital_id":"hosp-tmh",
  "eta_to_pickup_min":7.8, "eta_min":16.4, "distance_km":9.1,
  "traffic_factor":1.3, "track_token":"ab12...", "created_at":"..." }
```

6. ShuttleCards

Employee shuttle entitlement cards and their ride history (secondary module sharing the platform).

6.1 Schema & access

Attribute / pattern	Notes
PK = CARD#<id>, SK = META	Card record: holder, grade, monthly cap, used_this_month
SK = RIDE#<date>#<id>	A ride row under the card
Scan / Query PK	List cards; list a card's rides (SK begins_with RIDE#)

7. Cross-table notes

- Reference data is cached in-Lambda per warm invocation to keep dispatch fast.
- Bed capacity is intentionally NOT tracked here — hospital systems own it.
- Audit/event rows (EVT#) are append-only for traceability.
- On-demand billing means tables cost ~nothing at idle and scale with traffic.